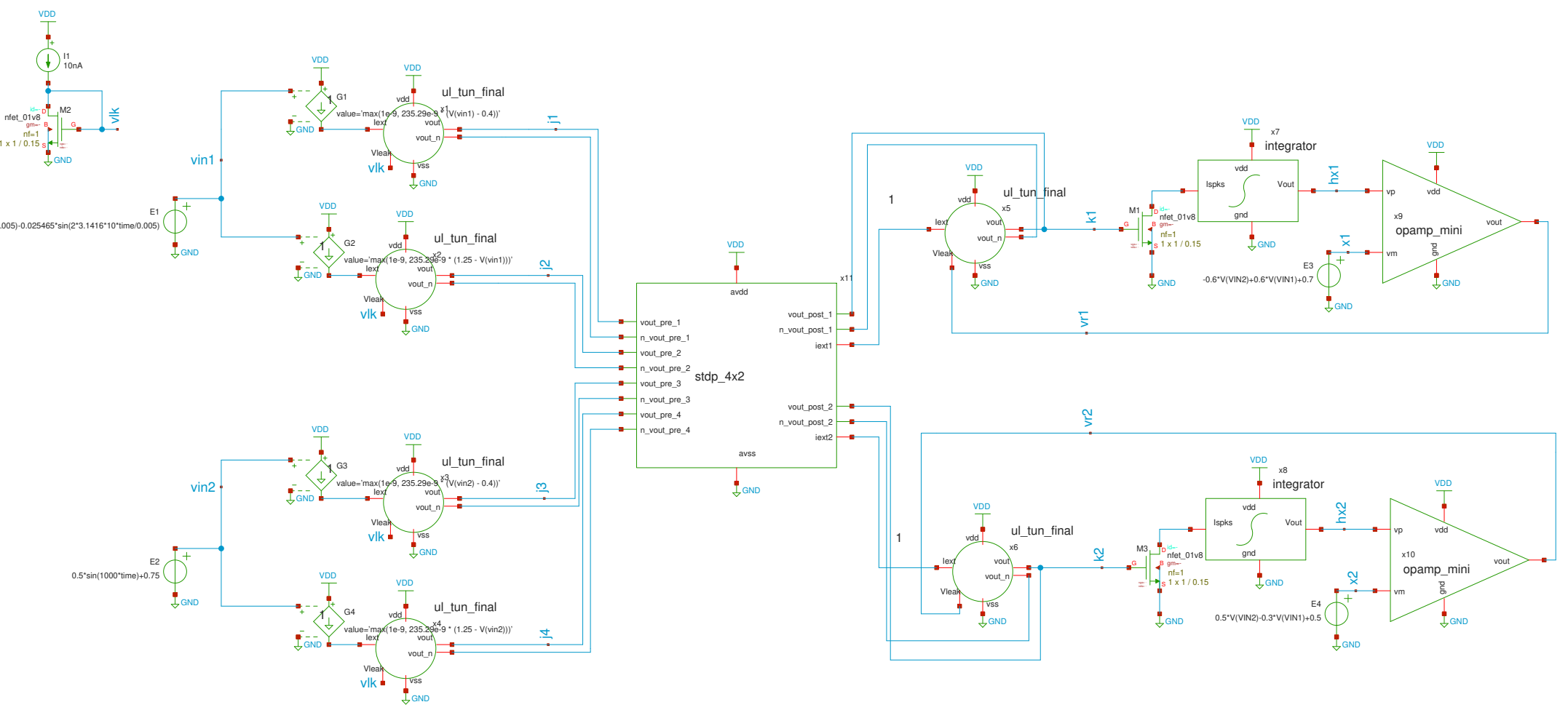
VDD
I1
10nA
M2
nfet_01v8
nf=1
1 x 1 / 0.15
vlk
GND
vin1
E1
.005)-0.025465*sin(2*3.1416*10*time/0.005)
G1
value='max(1e-9, 235.29e-9 * (V(vin1) - 0.4))'
G2
value='max(1e-9, 235.29e-9 * (1.25 - V(vin1)))'
ul_tun_final
x1
x2
vdd
vout
vout_n
Vleak
vss
j1
j2
vin2
E2
0.5*sin(1000*time)+0.75
G3
value='max(1e-9, 235.29e-9 * (V(vin2) - 0.4))'
G4
value='max(1e-9, 235.29e-9 * (1.25 - V(vin2)))'
x3
x4
j3
j4
stdp_4x2
avdd
avss
vout_pre_1
n_vout_pre_1
vout_pre_2
n_vout_pre_2
vout_pre_3
n_vout_pre_3
vout_pre_4
n_vout_pre_4
vout_post_1
n_vout_post_1
iext1
vout_post_2
n_vout_post_2
iext2
x11
1
k1
k2
vr1
vr2
M1
M3
x5
x6
integrator
x7
x8
lspks
Vout
hx1
hx2
opamp_mini
x9
x10
vp
vm
E3
-0.6*V(VIN2)+0.6*V(VIN1)+0.7
E4
0.5*V(VIN2)-0.3*V(VIN1)+0.5